

1) Soln:-

The Graph is on $[0, 6]$ so,

$$\Delta x = \frac{6-0}{6} = 1,$$

each rectangle has width 1.

a) Left-endpoint Riemann sum

The left endpoints are

$$x = 0, 1, 2, 3, 4, 5$$

from the graph

x	$f(x)$
0	2
1	3.5
2	4
3	2
4	-0.5
5	-3

Therefore

$$L_6 = \sum_{i=0}^5 f(x_i) \Delta x$$

$$= (2 + 3.5 + 4 + 2 - 0.5 - 3)(1)$$

$$= 8$$

classmate
Date _____
Page _____

b) Midpoint Riemann Sum

The midpoints are

0.5, 1.5, 2.5, 3.5, 4.5, 5.5

Reading the graph:

Midpoint	$f(x)$
0.5	2.8
1.5	3.8
2.5	3.6
3.5	0.8
4.5	-2.1
5.5	-3.2

Thus,

$$M_6 = (2.8 + 3.8 + 3.6 + 0.8 - 2.1 - 3.2)(1) \\ = 5.7$$

3) Evaluate $\int_0^1 (n + \sqrt{1-n^2}) dn$ by interpreting it in terms of areas.

Soln:-

$$\text{Let } A = \int_0^1 (n + \sqrt{1-n^2}) dn$$

$$\text{or, } A = \int_0^1 n dn + \int_0^1 \sqrt{1-n^2} dn$$

$$\text{or } A = \left[\frac{n^2}{2} \right]_0^1 + \int_0^1 \sqrt{1-n^2} dn$$

The second integral $\int_0^1 \sqrt{1-n^2} dn$ represents the

area $y = \sqrt{1-n^2}$ which represents the upper half of the circle $n^2 + y^2 = 1$ which is a unit circle. From $n=0$ to $n=1$, the region is a quarter of a unit circle. So,

$$\int_0^1 \sqrt{1-n^2} dn = \frac{1}{4} \pi (1)^2 = \frac{\pi}{4}$$

Now,

$$\text{or, } A = \left[\frac{1}{2} - \frac{0}{2} \right] + \frac{\pi}{4}$$

$$\text{or } A = \frac{1}{2} + \frac{\pi}{4}$$

5) If $\int_0^6 f(x) dx = 10$ and $\int_0^4 f(x) dx = 7$. find

$$\int_4^6 f(x) dx$$

Soln:-

Given;

$$\int_0^6 f(x) dx = 10$$

$$\int_0^4 f(x) dx = 7$$

Using the property of definite integral,

$$\int_0^4 f(x) dx + \int_4^6 f(x) dx = \int_0^6 f(x) dx$$

$$\text{or, } 7 + \int_4^6 f(x) dx = 10$$

$$\text{or, } \int_4^6 f(x) dx = 10 - 7$$

$$\text{or, } \int_4^6 f(x) dx = 3$$

$$\therefore \int_4^6 f(x) dx = 3$$

7) Soln:-

Let

$$F(x) = \int_0^x f(t) dt$$

Using the Fundamental Theorem of Calculus,

$$F'(x) = f(x)$$

So:

- F must pass through $(0,0)$.
- Where f has a maximum or minimum, $f=0$.
- Where f has a maximum or minimum $f'=0$.

Finding F

Graphs a and b pass through origin, but graph c does not. Since

$$F(0) = \int_0^0 f(t) dt = 0,$$

graph c cannot be F .

In the graph b, there has a local maximum exactly where graph a crosses the x-axis. It also has a local minimum where graph a crosses again.

Thus a is the derivative of b.

Therefore:

$$b = \int_0^a f(t) dt$$

and,

$$a = f$$

Identifying f'

Since $a = f$, the remaining graph must be f' .

Graph a has ~~min~~ minimum where graph c crosses the x -axis and graph a has a maximum where graph c crosses again. That matches the derivative relationship.

Hence,

$$c = f'$$

$$\text{So, } a = f, \quad b = \int_0^a f(t) dt, \quad c = f'$$

9) Soln:-

Given,

$$g(n) = \int_0^n f(t) dt$$

From the graph, There is given,

- $(0, -2)$ to $(2, 2)$
- horizontal from $n=2$ to $n=3$ at $y=2$
- then decreasing from $(3, 2)$ to $(5, -2)$

To find $g(4)$

Compute the signed area from 0 to 4.

From 0 to 2

The line crosses the x-axis at $n=1$.The triangle below axis on $[0, 1]$ is

$$-\frac{1}{2} (1)(2) = -1$$

The triangle above axis on $[1, 2]$ is

$$+\frac{1}{2} (1)(2) = +1$$

So, the net area is 0

From range 2 to 3

Rectangle is
 $(1)(2) = 2$

from the range 3 to 4

Triangle with base 1, height 2

$$\frac{1}{2} (2)(1) = 1$$

So,

$$g(4) = 0 + 2 + 1 = 3$$

$$\therefore g(4) = 3$$

To find $g'(4)$

Using the fundamental theorem of calculus

$$g'(x) = f(x)$$

At $x=4$, the graph of f crosses the x -axis
So,

$$f(4) = 0$$

$$\therefore g'(4) = 0$$

11-42) Evaluate the integral, if it exists.

$$1) \int_{-1}^0 (n^2 + 5n) \cdot dn$$

Soln:-

$$\text{Let } I = \int_{-1}^0 (n^2 + 5n) dn$$

$$\text{or } I = \int_{-1}^0 n^2 dn + \int_{-1}^0 5n dn$$

$$\text{or } I = \left[\frac{n^3}{3} \right]_{-1}^0 + 5 \left[\frac{n^2}{2} \right]_{-1}^0$$

$$\text{or } I = \frac{1}{3} [0^3 - (-1)^3] + \frac{5}{2} [0^2 - (-1)^2]$$

$$\text{or } I = \frac{1}{3} [0 + 1] + \frac{5}{2} [0 - 1]$$

$$\text{or } I = \frac{1}{3} - \frac{5}{2}$$

$$\text{or } I = \frac{2 \times 1 - 5 \times 3}{6}$$

$$\text{or } I = \frac{-13}{6}$$

$$13) \int_0^1 (1-x^9) dx$$

Soln:-

$$\text{Let } I = \int_0^1 (1-x^9) dx$$

$$\text{or } I = \int_0^1 dx - \int_0^1 x^9 dx$$

$$\text{or } I = [x]_0^1 - \left[\frac{x^{10}}{10} \right]_0^1$$

$$\text{or } I = 1 - 0 - \left[\frac{1^{10}}{10} - \frac{0^{10}}{10} \right]$$

$$\text{or } I = 1 - \frac{1}{10}$$

$$\text{or } I = \frac{10 \times 1 - 1}{10}$$

$$\text{or } I = \frac{9}{10}$$

$$15) \int_1^9 \frac{\sqrt{u} - 2u^2}{u} du$$

Soln:-

$$\text{Let } I = \int_1^9 \frac{\sqrt{u} - 2u^2}{u} du$$

$$\text{cn } I = \int_1^9 \frac{u^{\frac{1}{2}}}{u} - 2 \int_1^9 u^{2-1} du$$

$$\text{cn } I = \int_1^9 u^{\frac{1}{2}-1} du - 2 \int_1^9 u du$$

$$\text{cn } I = \int_1^9 u^{-\frac{1}{2}} du - 2 \left[\frac{u^2}{2} \right]_1^9$$

$$Q_7 \quad I = \left[\frac{4^{\frac{1}{2}+1}}{\frac{-1}{2}+1} \right]_1^9 - [81-1]$$

$$Q_7 \quad I = \left[\frac{4^{\frac{1}{2}}}{\frac{1}{2}} \right]_1^9 - 80$$

$$Q_7 \quad I = 2[\sqrt{4}]_1^9 - 80$$

$$Q_7 \quad I = 2[3-1] - 80$$

$$Q_7 \quad I = 2 \times 2 - 80$$

$$Q_7 \quad I = 4 - 80$$

$$Q_7 \quad I = -76 //$$

$$12) \int_0^1 y(y^2+1)^5 dy$$

Soln:-

$$\text{Let } I = \int_0^1 y(y^2+1)^5 dy$$

$$\text{Put } u = y^2+1$$

$$\text{or, } \frac{du}{dy} = \frac{d(y^2+1)}{dy}$$

$$\text{when } y=1, u=2$$

$$\text{when } y=0, u=1$$

$$\text{or } du = 2y \cdot dy$$

Now,

$$I = \frac{1}{2} \int_1^2 u^5 du$$

$$I = \frac{1}{2} \left[\frac{u^6}{6} \right]_1^2$$

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Date

Page

$$I = \frac{1}{12} [2^6 - 1]$$

$$I = \frac{63}{12}$$

$$I = \frac{21}{4}$$

$$19) \int_1^5 \frac{dt}{(t-4)^2}$$

Soln:-

$$\text{Let } I = \int_1^5 \frac{dt}{(t-4)^2}$$

$$\text{or } I = \int_{-1}^5 (t-4)^{-2} dt$$

$$\text{or } I = \left[\frac{(t-4)^{-2+1}}{(-2+1) \cdot 1} \right]_{-1}^5$$

$$\text{or } I = \frac{(t-4)^{-1}}{-1}$$

$$\text{or } I = \left[-\frac{1}{t-4} \right]_{-1}^5$$

However, at $t=4$, 4 lies inside the interval $[1, 5]$ and the integrand $\frac{1}{(t-4)^2}$ is undefined

at $t=4$.

So, this is an improper integral:-

$$\int_1^5 \frac{dt}{(t-4)^2} = \int_1^4 \frac{dt}{(t-4)^2} + \int_4^5 \frac{dt}{(t-4)^2}$$

Evaluating the limits,

$$\int_1^4 \frac{dt}{(t-4)^2} = \lim_{a \rightarrow 4^-} \left[-\frac{1}{t-4} \right]_1^a = \infty$$

and,

$$\int_4^5 \frac{dt}{(t-4)^2} = \lim_{b \rightarrow 4^+} \left[-\frac{1}{t-4} \right]_b^5 = \infty$$

So,

$$I = \int_1^5 \frac{dt}{(t-4)^2} = \infty \text{ which doesn't exist}$$

$$2) \int_0^1 v^2 \cos(v^3) dv$$

Soln:-

$$\text{Let } I = \int_0^1 v^2 \cos(v^3) dv$$

$$\text{Let } v^3 = u$$

$$\text{or } \frac{du}{dv} = 3v^2$$

$$\text{or } du = 3v^2 dv$$

$$\text{When } v=1, u=1$$

$$\text{When } v=0, u=0$$

Now,

$$\text{or } I = \frac{1}{3} \int_0^1 \cos u \cdot du$$

$$\text{or } I = \frac{1}{3} [\sin u]_0^1$$

$$\text{or } I = \frac{1}{3} [\sin 1 - \sin 0]$$

$$\text{or } I = \frac{1}{3} \sin 1$$

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$$23) \int_{-\pi/4}^{\pi/4} \frac{t^4 \tan t}{2 + \cos t} \cdot dt$$

Soln:-

$$\text{Let } I = \int_{-\pi/4}^{\pi/4} \frac{t^4 \tan t}{2 + \cos t} \cdot dt$$

$$\text{Put } f(t) = f(-t)$$

$$f(-t) = \frac{(-t)^4 \tan(-t)}{2 + \cos(-t)}$$

We know,

$$(-t)^4 = t^4,$$

$$\tan(-t) = -\tan t$$

$$\cos(-t) = \cos t$$

$$\text{We get, } f(-t) = \frac{t^4(-\tan t)}{2 + \cos t} = -\frac{t^4 \tan t}{2 + \cos t}$$

So,

$$f(-t) = -f(t)$$

The given function is odd function.

$$\therefore I = 0$$

$$25) \int \frac{x}{x^2+1} dx$$

Soln:-

$$\text{Let } I = \int \frac{x}{x^2+1} dx$$

$$\text{or, } I = \frac{1}{2} \int \frac{2x}{x^2+1} dx$$

$$\text{on } I = \frac{1}{2} \ln|x^2+1| + C$$

$$27) \int \frac{x+2}{\sqrt{x^2+4x}} dx$$

Soln:-

$$\text{Let } I = \int \frac{x+2}{\sqrt{x^2+4x}} dx$$

$$\text{Put } x^2+4x = u$$

$$\text{or, } \frac{du}{dx} = \frac{d(x^2+4x)}{dx}$$

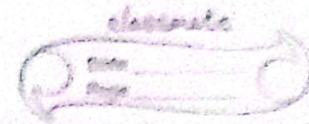
$$\text{on } \frac{du}{dx} = \frac{dx^2}{dx} + 4 \frac{dx}{dx}$$

$$\text{on } \frac{du}{dx} = 2x+4$$

$$\text{on } du = (2x+4) dx$$

Now,

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$$\text{or } I = \frac{1}{2} \int \frac{2n+4}{\sqrt{n^2+4n}} \cdot dn$$

$$\text{or } I = \frac{1}{2} \int \frac{du}{\sqrt{u}}$$

$$\text{or } I = \frac{1}{2} \int u^{-\frac{1}{2}} du$$

$$\text{or } I = \frac{1}{2} \left[\frac{u^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} \right] + C$$

$$\text{or } I = \frac{1}{2} \left[\frac{u^{\frac{1}{2}}}{\frac{1}{2}} \right] + C$$

$$\text{or } I = \frac{2}{2} \sqrt{u} + C$$

$$\text{or } I = \sqrt{n^2+4n} + C$$

$$29) \int \sin \pi t \cos \pi t \cdot dt$$

Soln:-

$$\text{let } I = \int \sin \pi t \cos \pi t \cdot dt$$

$$\text{Put } u = \sin \pi t$$

$$\text{on } \frac{du}{dt} = \frac{d \sin \pi t}{d \pi t} \cdot \frac{d \pi t}{dt}$$

$$\text{on } \frac{du}{dt} = \pi \cos \pi t$$

$$\text{on } du = \pi \cos \pi t \cdot dt$$

Now,

$$\text{on } I = \frac{1}{\pi} \int \pi \sin \pi t \cos \pi t \cdot dt$$

$$\text{on } I = \frac{1}{\pi} \int u \, du$$

$$\text{on } I = \frac{1}{\pi} \left[\frac{u^2}{2} \right] + C$$

$$\text{on } I = \frac{1}{2\pi} \sin^2 \pi t + C$$

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classmate

Date
Page

$$3) \int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$$

Solⁿ:-

$$\text{let } I = \int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$$

$$\text{Put } \sqrt{x} = y$$
$$\text{on } \frac{dy}{dx} = \frac{1}{2\sqrt{x}}$$

$$\text{on } \frac{dy}{dx} = \frac{1}{2} x^{\frac{1}{2} - 1}$$

$$\text{on } \frac{dy}{dx} = \frac{1}{2} x^{-\frac{1}{2}}$$

$$\text{on } \frac{dy}{dx} = \frac{1}{2\sqrt{x}}$$

$$\text{on } dy = \frac{1}{2\sqrt{x}} dx$$

$$\text{on } I = \frac{1}{2} \int \frac{2 e^{\sqrt{x}}}{\sqrt{x}} dx$$

$$\text{on } I = 2 \int \frac{e^{\sqrt{x}}}{2\sqrt{x}} dx$$

$$\text{on } I = 2 \int e^y dy$$

$$\text{ex } I = 2e^y + C$$

$$\text{ex } I = 2e^{\sqrt{x}} + C$$

$$33) \int \tan x \ln(\cos x) dx$$

Soln:-

$$\text{let } I = \int \tan x \ln(\cos x) dx$$

$$\text{let } u = \ln(\cos x)$$

$$\text{ex } \frac{du}{dx} = \frac{d \ln(\cos x)}{d \cos x} \cdot \frac{d \cos x}{dx}$$

$$\text{ex } \frac{du}{dx} = \frac{1}{\cos x} \cdot -\sin x$$

$$\text{ex } du = -\tan x dx$$

Now,

$$\text{ex } I = - \int -\tan x \ln(\cos x) dx$$

$$\text{ex } I = - \int u dx$$

$$\text{ex } I = - \left[\frac{u^2}{2} \right] + C$$

$$\text{ex } I = - \frac{1}{2} [\ln(\cos x)]^2 + C$$

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35) $\int \frac{x^3}{1+x^4} \cdot dx$

Soln:-

Let $I = \int \frac{x^3}{1+x^4} \cdot dx$

Let $u = 1+x^4$

on $\frac{du}{dx} = 4x^3$

on $du = 4x^3 dx$

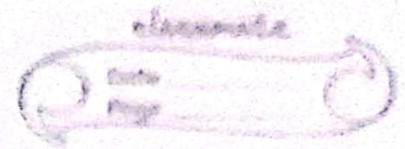
Now,
or, $I = \frac{1}{4} \int \frac{4x^3}{1+x^4} \cdot dx$

or, $I = \frac{1}{4} \int \frac{du}{u}$

or, $I = \frac{1}{4} \ln|u| + C$

or, $I = \frac{1}{4} \ln|1+x^4| + C$

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$$32) \int \frac{\sec \theta \cdot \tan \theta}{1 + \sec \theta} d\theta$$

Soln:

$$\text{Let } I = \int \frac{\sec \theta \cdot \tan \theta}{1 + \sec \theta} d\theta$$

$$\text{Put } u = 1 + \sec \theta$$

$$\text{or } \frac{du}{d\theta} = \sec \theta \cdot \tan \theta$$

$$\text{or } du = \sec \theta \cdot \tan \theta \cdot d\theta$$

$$\therefore I = \int \frac{du}{u}$$

$$\therefore I = \ln|u| + C$$

$$\therefore I = \ln|\sec \theta + 1| + C =$$

39) $\int x(1-x)^{2/3} \cdot dx$

Soln:-

let $I = \int x(1-x)^{2/3} \cdot dx$

Put $u = 1-x$

or, $\frac{du}{dx} = \frac{d(1-x)}{dx}$

or $\frac{du}{dx} = 0-1$

or $du = -dx$

or $dx = -du$

$u = 1-x$

$u+x = 1$

$x = 1-u$

Now,
or $I = \int (1-u) u^{2/3} (-du)$

or $I = \int u^{2/3} - u^{1+\frac{2}{3}} \cdot (-du)$

or $I = \int u^{2/3} - u^{5/3} du$

or $I = - \left[\frac{u^{\frac{2}{3}+1}}{\frac{2}{3}+1} \right] - \left[\frac{u^{\frac{5}{3}+1}}{\frac{5}{3}+1} \right] + C$

or $I = - \left[\frac{u^{5/3}}{5/3} \right] - \left[\frac{u^{8/3}}{8/3} \right] + C$

Shiny leaf

$$\text{or, } I = -\frac{3}{5} u^{5/3} - \frac{3}{8} u^{8/3} + C$$

$$\text{or } I = -\frac{3}{5} u^{5/3} + \frac{3}{8} u^{8/3} + C$$

$$\text{or } I = -\frac{3}{5} (1-x)^{5/3} + \frac{3}{8} (1-x)^{8/3} + C$$